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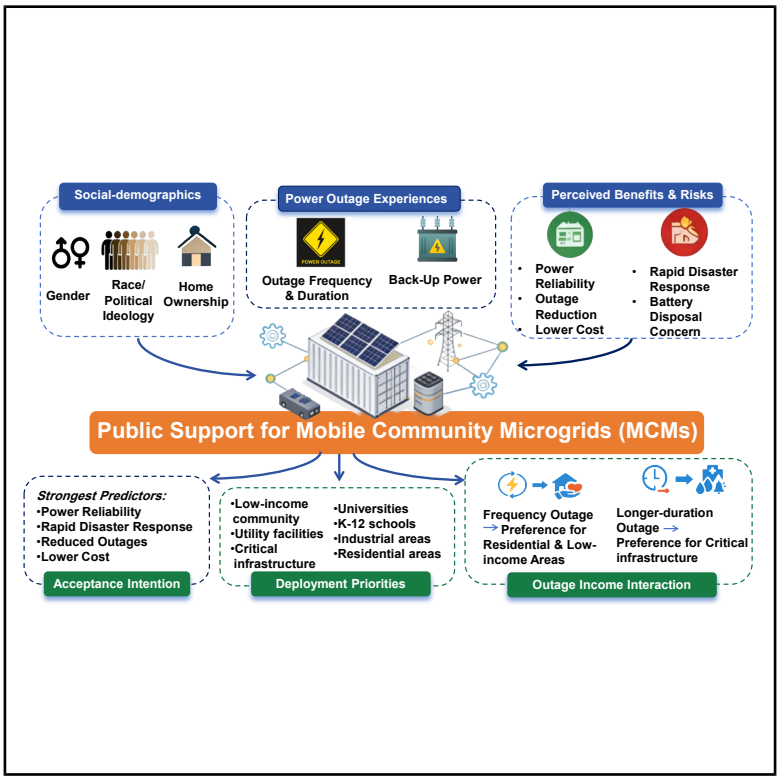
Public support for mobile community microgrids: Socioeconomic, perceptual and outage experience determinants of energy resilience

Graphical abstract



Highlights

- Perceptions shape acceptance of mobile community microgrids (MCMs)
- More frequent power outages are linked to higher MCM acceptance
- Higher perceived MCM reliability is linked to higher MCM acceptance
- Faster MCM disaster response increases acceptance

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In brief

Energy resources; energy policy; energy sustainability

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Public support for mobile community microgrids: Socioeconomic, perceptual and outage experience determinants of energy resilience

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SUMMARY

As climate-induced disasters increasingly compromise electric grid reliability, mobile community microgrids (MCMs) have emerged as a promising strategy to strengthen local energy resilience. This study examines the socioeconomic and perceptual determinants of public MCM acceptance using a nationally representative survey of 1,996 U.S. residents. Hierarchical regression results indicate that the desire for improved power reliability is the strongest predictor of acceptance, followed by expectations of faster disaster response and lower energy costs. Power outage experience is also a significant driver of support, with stronger effects among men and respondents who face frequent disruptions. Although political ideology appears influential in baseline models, its association attenuates after accounting for economic conditions. Preferences for deployment locations further diverge by outage experience: respondents with frequent outages prioritize residential and disadvantaged communities, whereas those with fewer disruptions place greater emphasis on critical infrastructure. These findings highlight the need for MCM deployment strategies that are responsive to heterogeneous community perceptions and place-based resilience priorities.

INTRODUCTION

The increasing frequency and severity of climate-related disasters, including hurricanes, wildfires, ice storms, and heatwaves, have placed mounting stress on the United States' aging, centralized electric grid.[1,2] High-profile outage events, including the 2021 Texas winter storm that left some communities without power for over 10 days,[3,4] and the widespread, prolonged shutoffs during the 2019–2020 California wildfire seasons,[5] underscore the grid's vulnerability to extreme weather and systemic failure. These outages disproportionately affect low-income households, medically vulnerable individuals, and underserved neighborhoods, exacerbating existing social and health inequities.[6–9] In response, microgrids have gained attention as resilient infrastructure solutions.[10]

Microgrids integrate distributed energy resources, storage systems, loads, and control components into a unified, controllable power-supply system.[11] Capable of operating independently from the main grid,[12] microgrids are proven effective at bolstering local energy resilience,[13] especially in supporting critical infrastructure during power outages and enabling renewable energy integration.[14] However, stationary microgrids are often constrained by high installation costs, regulatory complexity, and lengthy implementation timelines, limiting their utility in disaster situations.[15] To address these limitations, mobile community microgrids (MCMs) have emerged as scalable, rapidly deployable alternatives designed to deliver reliable, temporary electricity during climate-related disruptions.[16–18] MCMs are small-scale local electric networks that are movable, integrating renewable energy (e.g., solar panels), battery storage, and backup generators in shipping containers. The value of mobile energy systems was evident following Hurricane Sandy, when more than 400 truck-mounted generators were deployed to restore emergency power across the northeastern U.S.[19,20] Despite their short-term success, these diesel-powered units relied on fragile regional fuel supply chains, leading to severe shortages and rationing in New York and New Jersey.[21] Their high emissions and standalone functionality[22] further revealed logistical and environmental drawbacks, underscoring the need for cleaner, integrated mobile energy solutions such as MCMs that align with long-term resilience and sustainability objectives.

Microgrids, including MCMs, have demonstrated significant potential to reduce the disproportionate impacts of power outages on vulnerable populations.[23] Recent studies have begun to highlight the distinctive advantages of mobile microgrids in enhancing energy resilience. Unlike stationary microgrids, MCMs can be rapidly deployed to restore power after extreme events, support emergency operations, or provide temporary generation in damaged grid areas. Their modularity and

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transportability enable fast actions to power outages and scalable support for critical facilities and underserved communities. For instance, mobile hybrid microgrids are technically[24] and economically[25] feasible for rapid disaster response and improving energy resilience; mobile multi-energy microgrids are capable of dynamic community applications from serving highly dispersed, seasonally migrating off-grid settlements to reconfigurable, grid-connected clusters of communities.[25] Together, these studies demonstrate that MCMs not only provide resilience and sustainability benefits as stationary microgrids, but also add flexibility of mobility, making them an emerging solution for both emergency response and equitable energy access.

Recent system-level studies propose flexible and mobile microgrid concepts to enhance reliability and resilience through advanced coordination and optimization. Zhou introduces spatiotemporal microgrids that incentivize multi-stakeholder participation in cross-boundary energy sharing,[26] while Zhou and Lund review peer-to-peer trading models in smart communities that enable more dynamic and decentralized energy management.[27] Building on this line of work, Zhou et al. show how stochastic vehicle scheduling in renewable-building-e-transportation microgrids can provide demand response and grid flexibility,[28] and Dan et al. develop city information models that integrate electric vehicle charging and distributed resources to support energy-resilient urban infrastructure.[29] Our findings complement these technical and planning-oriented contributions by demonstrating that the success of MCMs also depends on social acceptance, trust in implementing institutions, and perceptions of equity in how resilient resources are allocated.

Despite advances in microgrid engineering, the socio-economic and social-psychological dimensions of MCM acceptance remain underexplored.[11] Much of the existing literature centers on technical design, grid integration, and performance metrics such as reliability,[30] and operational efficiency[30,31] and environmental impacts,[32] often neglecting equity consideration.[33–36] While resilience concerns and prior experience with extreme weather have been identified as key motivators for microgrid acceptance,[37,38] research on how these experiences shape public interest remains limited.[39]

Empirical findings are mixed, leaving key questions unanswered: Which populations are most receptive to MCM technology? What social-psychological factors drive or hinder acceptance? And how do past outage experiences influence support for microgrid deployment? As with other energy innovations, successful diffusion of MCMs depends not only on technical viability but also on public trust, perceived benefits and risks, and community alignment.[40] Increasingly, energy justice and procedural equity have become central to infrastructure planning.[41,42] Yet resilience strategies are often implemented through top-down approaches[43,44] that overlook the lived experiences of those most affected by energy insecurity.[45,46] To ensure MCM investments promote community resilience, it is critical to understand how socio-demographic characteristics and social-psychological factors, such as perceived reliability, cost-effectiveness, and fairness, influence public support and preferred deployment locations.[47]

Purpose of this study

While most research on MCMs emphasizes technological design and performance aspects,[48,49] few studies have examined public acceptance and resilience-related dimensions. This study addresses that gap by exploring five critical dimensions of community acceptance: (1) socio-demographic characteristics, (2) perceived benefits and risks, (3) experience with power outages, (4) perceptions of power reliability, and (5) preferences for microgrid deployment sites. Based on our nationally representative survey of nearly 2,000 U.S. residents, this research aims to identify key drivers and barriers to MCM acceptance. Specifically, the study investigates the following research questions (*RQs*): *RQ1:* How are sociodemographic factors associated with residents' power-outage experiences, such as outage frequency and duration? *RQ2:* Which sociodemographic characteristics and outage experiences predict ownership of backup power systems? *RQ3:* What sociodemographic characteristics are associated with higher intentions to adopt MCMs? *RQ4:* To what extent do prior power outage experiences influence willingness to adopt MCMs? *RQ5:* How do perceived benefits and risks relate to power reliability, safety, and financial cost shape acceptance of MCMs? *RQ6:* What deployment locations do residents prioritize for MCM installation?

This research uses the integrated framework that weaves the Technology Acceptance Model (TAM),[50,51] Protection Motivation Theory (PMT),[52,53] and the Energy Justice framework to interpret public acceptance of MCM and to guide equitable and community-informed deployment. TAM explains how perceived usefulness and ease of use shape intentions to adopt technologies, including smart meters, renewable energy, and solar panel and microgrid acceptance.[50,51,54] PMT clarifies why people take protective actions when they perceive a threat[53,55]; protection motivation increases when perceived severity and vulnerability are high, maladaptive rewards are low, and both efficacy beliefs are high while response costs are manageable. In the context of microgrids, PMT links how outage severity and perceived vulnerability combine with beliefs about response efficacy, self-efficacy, and response costs to explain intentions to adopt MCMs. Households are therefore most likely to accept microgrids when outages feel serious and likely, they believe MCMs will meaningfully improve reliability, they feel confident they can access the system, and financial and practical barriers are reduced.

The Energy Justice framework provides a structural lens and decision-making tool for researchers and policy makers to evaluate whether benefits and burdens are fairly distributed, whether communities have a voice and transparency in decision-making, and whether histories, identities, and place-based harms are acknowledged, often alongside restorative justice aims.[56] These perspectives predict higher uptake when households face salient risks, view MCMs as effective and low-friction solutions, and participate in equitable, co-designed processes that build trust.[40,57] This study also probes distributional justice by examining siting preferences, for example, whether respondents prioritize locating MCMs in low-income neighborhoods and how perceived fairness shapes MCM support.

Figure 1 presents the study's conceptual framework. Using hierarchical regression, we estimate how socio-demographics,

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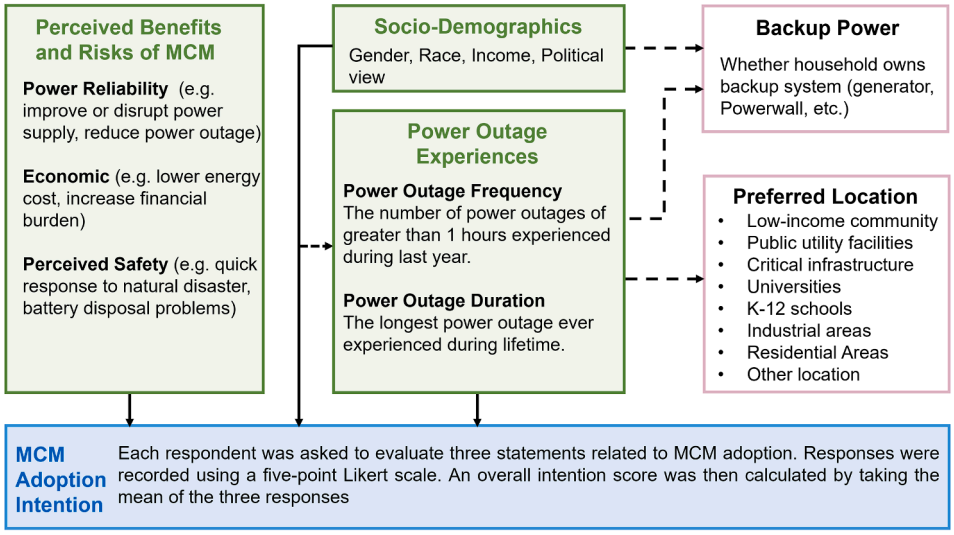

Figure 1. MCM acceptance framework

Solid black arrows indicate the primary analysis using a hierarchical linear regression. Dashed arrows indicate ancillary analyses examining (1) sociodemographic and outage experience, (2) sociodemographic and backup-power ownership, and (3) outage experience and preferred MCM deployment locations.

social-psychological factors, and power-resilience experiences incrementally explain variation in acceptance and support for MCMs. Table 1 summarizes the corresponding hypotheses.

RESULTS

The disparity of power resilience across income groups

To address *RQ1* (How are sociodemographic factors associated with residents' power-outage experiences, such as outage frequency and duration?), results of our analysis reveal a clear pattern of variation in power resilience experiences across income groups. Specifically, power outage experiences in terms of frequency and duration varied among three consolidated income levels: low income (less than $50,000 annually), middle income ($50,000 to $124,999), and high income (over $125,000). To enhance clarity in descriptive visualizations, we combined the original seven income categories into these three broader groups. However, in subsequent inferential analyses such as ANOVA, we used the full seven income categories to capture more detailed patterns and to better examine the relationship between income and power outage experiences.

As shown in Figures 2A–2C, over 75% of respondents reported experiencing 0–2 times of long power outages in the past year. The average frequency of outages was relatively consistent across the three income groups, falling within a narrow range of 1–2 times, while more than 25% of respondents experienced outages lasting more than two days. To explore the relationship between income and power outage experiences, we conducted two separate simple linear regressions using ordinary least squares (OLS), with power outage frequency and duration as the dependent variables, respectively. Results suggest income had an insignificant impact on frequency but a positive effect on power outage duration significantly (R^2 = 0.003, β = 0.055, $p < 0.05$). Similarly, results of one-way analysis of variance (ANOVA, Table S3) on seven income groups showed that income was not a significant factor in the outage frequency but significantly affects the longest power outage duration. Specifically, when considering the longest outage duration experienced (Figures 2D–2F), a surprising income gradient emerges: the high-income group reported the longest average duration, followed by the middle-income group, with the low-income group reporting the shortest. The post-hoc test results (Table S8) demonstrate that the lowest level of income households (income < $25,000) reported significantly lower values for the longest single outage they experienced (*mean* = 4.68, *s.d.* = 1.88) compared to the highest income group (>$150,000, *mean* = 5.26, *s.d.* = 1.77, $p < 0.01$). Additional robust analyses are reported in the Supplementary Material (Tables S4–S7). The income-outage duration pattern may also reflect confounding factors rather than purely socioeconomic effects. In some regions, higher-income households are located in suburban or wooded areas with greater vegetation density,[58] older power lines,[59,60] or lower customer density,[61] all of which can prolong restoration. Grid topology and rural feeder configurations may further amplify these restoration delays.[62] Accounting for these confounding influences provides a more balanced interpretation and highlights the need for future studies that integrate spatial and infrastructural data to disentangle geographic extreme weather impacts from socioeconomic effects.

Analysis of energy reliability systems across income levels

To investigate patterns in energy reliability systems, operationalized as home backup generator ownership, across different income levels, we conducted a linear regression analysis. Surprisingly, home backup power ownership alone was not a significant predictor of intention to accept MCMs (R^2 = 0.000, β = 0.046, p = 0.347). Further models that controlled for power outage

Table 1. Proposed hypotheses

Dimensions	Hypothesis
Socio-demographics	(H1) Power outage experiences significantly differ by the income level.
Resilience experience	(H2) Resilience experience is positively related to household backup generator ownership.
Socio-demographics	(H3a) Higher-income households show greater MCM acceptance. (H3b) Liberal political views are positively related to higher MCM acceptance. (H3c) MCM acceptance differs by gender. (H3d) White residents have a higher intention to accept MCM than non-white residents.
Resilience experience	(H4a) Higher outage frequency in the past year is positively associated with MCM acceptance. (H4b) Longer outage duration is positively associated with MCM acceptance.
Power reliabilities	(H5a) Perceived increase in power supply of MCMs leads to higher MCM acceptance. (H5b) Perceived reduction in outage likelihood leads to higher MCM acceptance. (H5c) Greater perceived power disruption risk is negatively associated with MCM acceptance.
Perceived safety	(H5d) Perceived improvements in rapid disaster response capacity are positively related to MCM acceptance. (H5e) Concerns about battery disposal and environmental harm are negatively related to MCM acceptance.
Economic	(H5f) Perceived reduction in energy costs is positively associated with MCM acceptance. (H5g) Perceptions of increased financial burden are negatively associated with MCM acceptance.

experiences and socio-demographic characteristics similarly showed no evidence of a significant relationship between home backup generator ownership and MCM acceptance intention. Although owning a home backup generator does not directly influence MCM acceptance intention, understanding generator ownership remains relevant, as both backup generators and MCMs are intended to enhance household resilience against power disruptions (*RQ2*). Characterizing households likely to own backup systems can provide valuable insights for targeted MCM implementation strategies.

Descriptive analyses (Figure 3) revealed notable ownership trends. Higher-income households (Figure 3D) and respondents with more conservative political orientations (Figure 3B) exhibited a greater likelihood of owning backup generators. Chi-square tests confirmed a significant association between income level and generator ownership ($\chi^2 = 24.35$, $p < 0.001$), as well as between political ideology and ownership ($\chi^2 = 10.41$, $p = 0.034$). Additionally, ownership correlated positively with the frequency of prolonged power outages experienced in the past year. A chi-square test also revealed a significant relationship between outage frequency and generator ownership ($\chi^2 = 56.58$, $p < 0.001$). Ownership appeared evenly distributed across different durations of the longest power outage experienced ($\chi^2 = 13.13$, $p = 0.041$).

To examine these relationships more rigorously, we conducted an additional regression analysis, modeling home backup power ownership as a function of socio-demographics and power outage experiences. As expected, the coefficient of power outage frequency was significantly positive, while the coefficient of power outage duration was insignificant (Figure 4). Also illustrated in Figure 4, income emerged as a significant positive predictor ($\beta = 0.024$, $p < 0.001$), indicating that higher-income respondents were more likely to own home backup power systems. Political views also had a significant positive relationship ($\beta = 0.020$, $p < 0.05$), reflecting a higher likelihood of ownership among respondents with conservative orientations. Race (white vs. non-white), however, was not a significant predictor, suggesting that race/ethnicity did not notably influence backup power ownership decisions.

Results of sociodemographic and risk and benefit factors influencing mobile community microgrid acceptance

To address *RQ3* (What sociodemographic characteristics are associated with higher intentions to adopt MCMs?), we conducted a multiple regression analysis (Model 1 in Table 2). Results indicated that socio-demographic characteristics alone accounted for 2.3% of the variance in intention to adopt MCMs ($R^2 = 0.023$, $F(4, 1626) = 9.550$, $p < 0.001$). Specifically, individuals holding more liberal political views ($\beta = -0.840$; $p < 0.001$) and males ($\beta = 0.136$; $p < 0.001$) showed significantly higher acceptance intentions. Annual income and race did not emerge as statistically significant predictors.

In addressing *RQ4* (To what extent do prior power outage experiences influence willingness to adopt MCMs?), we expanded Model 1 by incorporating participants' resilience experiences (Table 2, Model 2). Results revealed a statistically significant model ($F(6, 1624) = 8.584$, $p < 0.001$), which explained 3.1% of the variance in MCM acceptance intention, an incremental improvement of 0.8% of R^2 over Model 1 (F-change (2, 1624) = 6.523, $p < 0.001$). Consistent with Model 1, political ideology and gender (male) remained significant predictors, while income and race continued to have no significant effect. Regarding power outage experiences, the frequency of prolonged outages over the past year was positively associated with acceptance intentions ($\beta = 0.036$; $p < 0.01$). However, the duration of the longest outage experienced was not significantly related to MCM acceptance intention.

To address *RQ5* (How do perceived benefits and risks related to power reliability, safety, and financial cost shape acceptance of MCMs?), we developed Model 3, integrating socio-demographics, power resilience experiences, and perceived risks and benefits associated with power-related aspects of MCMs. Model 3 explained 39.1% of the variance in acceptance

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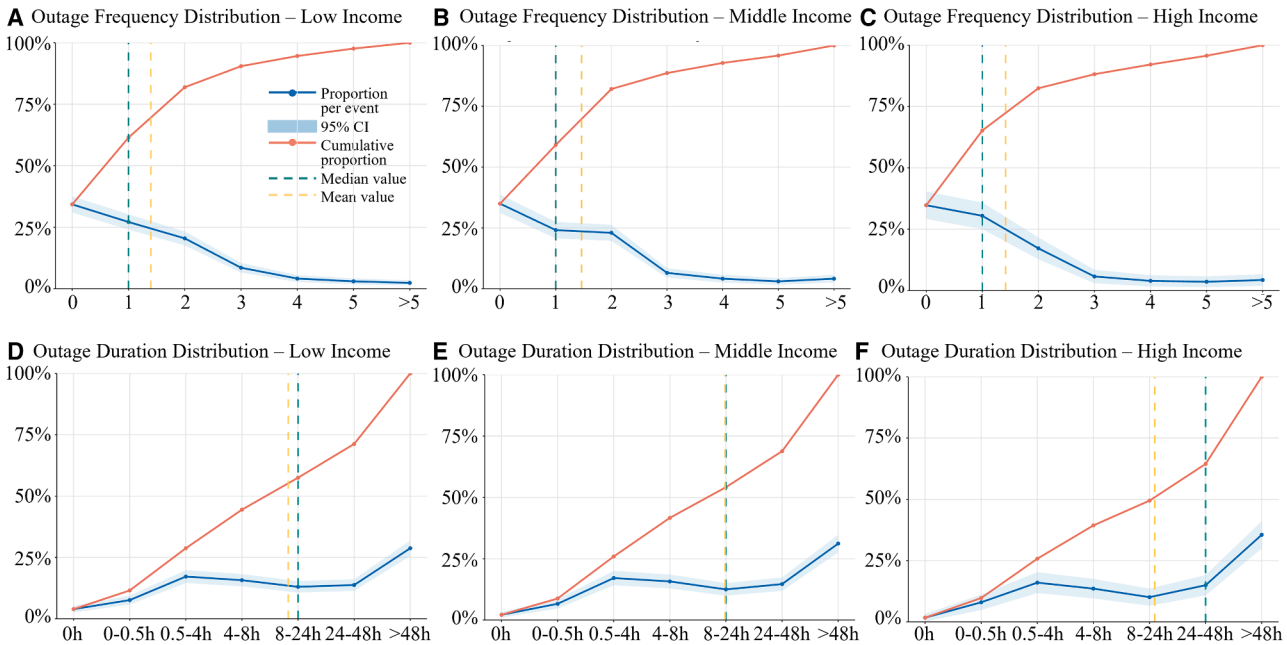

Figure 2. Power outage frequency and duration by income group

(A–C) depict the distribution of power outage frequency, defined as the number of times experiencing long power outages in the past year, across the low-, middle-, and high-income groups, respectively.

(D–F) illustrate the distribution of power outage duration, representing the longest outage experienced, across the same income categories. High-income individuals have longer outage durations on average, while outage frequency is more equally distributed among income groups.

intention, achieving an adjusted R^2 of 38.7%, (F(9, 1621) = 115.435, $p < 0.001$). Compared with Model 2, this represents a statistically significant increase in explanatory power, with an R^2 increment from 3.1% to 39.1% (F-change (3, 1621) = 319.05, $p < 0.001$). Consistent with previous findings, political views, gender, and power outage frequency remained significant predictors. Specifically, respondents perceiving MCMs as providing a more reliable power supply ($\beta = 0.505$, $p < 0.001$) and reducing outage occurrences ($\beta = 0.146$, $p < 0.001$) showed significantly higher acceptance intentions. Conversely, individuals who believed that MCMs would increase power supply disruptions exhibited significantly lower interest ($\beta = -0.057$, $p < 0.01$).

Model 4 further incorporated socio-demographic factors, power outage experiences, power-related perceptions, and safety-related perceptions regarding MCMs. This comprehensive model accounted for 42.4% of the variance in MCM acceptance intention, with an adjusted R^2 of 42.0% (F(11, 1619) = 108.186, $p < 0.001$). Compared to Model 3, the addition of safety-related perceptions significantly improved the explanatory power, evidenced by a statistically significant R^2 increase of 3.3% (F-change (2, 1619) = 46.44, $p < 0.001$). Political views, gender, and outage frequency remained significant predictors. Within power-related perceptions, beliefs about reliable power supply and outage reductions retained significance, whereas perceptions about increased power disruptions lost predictive significance when safety perceptions were considered. Among safety-related perceptions, only perceptions of rapid disaster responses emerged as a significant predictor ($\beta = 0.231$, $p < 0.001$), ranking as the second strongest predictor overall, while reliable power supply perception continued as the strongest predictor in Model 4.

Finally, Model 5 (Figure 5) incorporated finance-related perceptions concerning MCM acceptance, specifically expectations of lower energy costs and concerns about increased community financial burdens. Adding these financial variables produced a modest yet statistically significant improvement in the model, with an increase in R^2 from 42.4% to 43.8% (adjusted R^2 = 43.3%; F (13, 1617) = 96.80, $p < 0.001$). Specifically, perceived lower energy costs emerged as a significant positive predictor of acceptance intention ($\beta = 0.124$, $p < 0.001$), indicating that individuals anticipating reduced energy expenses were more inclined to adopt MCMs. Conversely, concerns regarding higher financial burdens did not significantly deter acceptance intention, as indicated by the non-significant coefficient for perceived financial burden.

The inclusion of community finance-related perceptions minimally influenced the significance of previously established predictors. The perception of reliable power supply remained the strongest predictor ($\beta = 0.349$, $p < 0.001$), followed by rapid disaster response ($\beta = 0.220$, $p < 0.001$), and reduced frequency of power outages ($\beta = 0.054$, $p < 0.05$). Notably, perceptions of increased power disruptions regained statistical significance ($\beta = -0.056$, $p < 0.01$) upon adding financial perceptions, whereas concerns about battery disposal issues remained non-significant. Among socio-demographic variables, political views lost significance in Model 5, suggesting that financial considerations may moderate the previously identified relationship

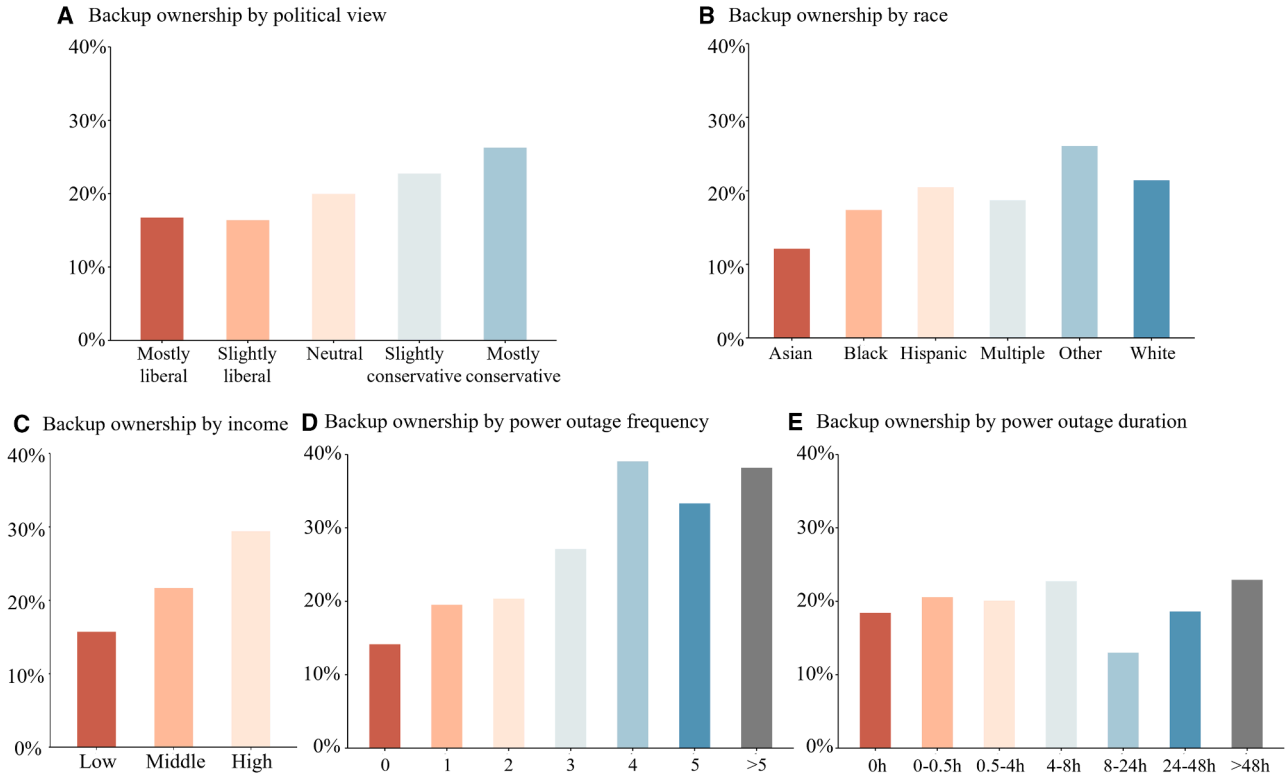

Figure 3. Proportion of respondents owning backup generators by socio-demographic, outage frequency, and duration
Each bar represents the percentage of individuals within a given subgroup who reported owning a generator.

between political orientation and MCM acceptance intention. Gender continued to be a significant predictor (β = 0.126, $p < 0.001$), while race and income consistently showed no significant effects.

Overall, these findings underscore the meaningful role of community financial incentives, particularly expectations of reduced energy costs, in encouraging MCM acceptance. Conversely, concerns about increased financial burdens do not appear influential. Thus, incorporating finance-related perceptions enhances the model's explanatory power without substantially altering the impact of previously identified core predictors.

Preferred locations for mobile community microgrids

Identifying optimal locations for microgrid deployment based on residents' perspectives is essential for advancing community-driven energy resilience. To answer *RQ6* (What deployment locations do residents prioritize for MCM installation?), we ask respondents to rank the preferred locations if their town/city is considering installing a microgrid. Respondents' choices not only reflect geographic preferences but also reveal which functions they believe MCMs should prioritize, such as protecting life-saving infrastructure, supporting vulnerable populations, or ensuring household continuity. Residents' lived experiences with outages, and their knowledge of local infrastructure, provide valuable guidance for siting decisions that maximize social impact. Aligning microgrid placement with actual community needs helps ensure that critical sites, such as residential neighborhoods, community centers, and essential public facilities, receive prioritized backup power. Such alignment fosters energy equity by incorporating the voices of those most affected by grid failures, ultimately supporting more responsive and effective deployment strategies.

Respondents expressed a clear hierarchy of preferences for MCM siting as shown in Figure 6. Critical infrastructure (e.g., hospitals, emergency operations centers, water and wastewater facilities) received the highest level of support, garnering nearly half of all responses. This reflects a strong public consensus on safeguarding life-sustaining services. Low-income communities ranked second, indicating that equity concerns are salient when the public evaluates resilience investments. A middle tier of support emerged for residential areas and public utility facilities, suggesting value placed on both individual household reliability and municipal service continuity. In contrast, universities, K-12 schools, and industrial areas received relatively limited support, pointing to lower perceived urgency for protecting these types of facilities compared to essential services or vulnerable populations. Overall, these patterns suggest that respondents weigh both societal benefit and distributive fairness when considering MCM deployment priorities.

To examine how power outage experiences shape location preferences, we constructed cross-tabulations (Figures 7A and 7B) comparing preferred MCM siting with outage frequency and duration. We calculated Pearson's chi-square statistics to assess the strength of associations and analyzed the relative

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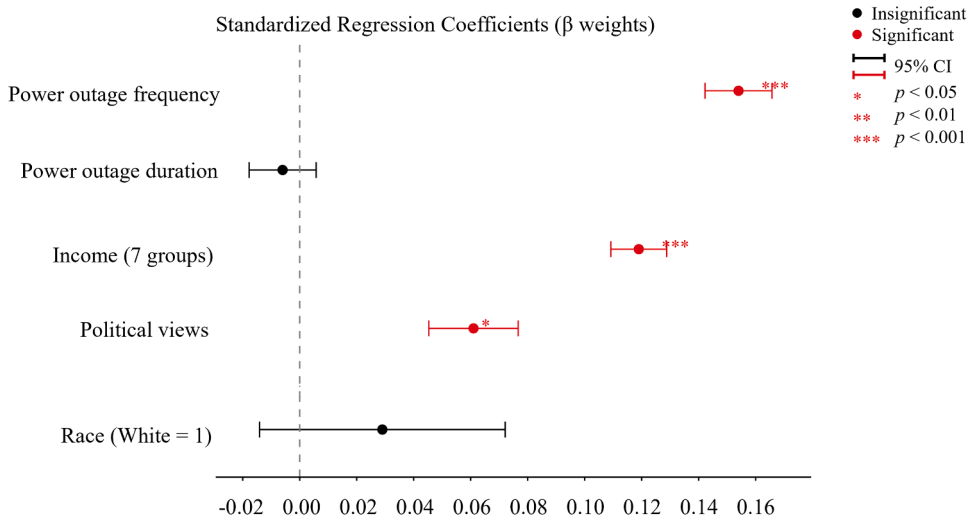

Figure 4. Estimated coefficients of linear regression for home backup generator ownership
Frequent outages, higher income, conservative orientation, and being male significantly predict backup generator ownership.

percentage change for each location type compared to its overall distribution. This approach allowed us to detect shifts in preference linked to different levels of outage exposure.

Results indicate a significant relationship between outage frequency and preferred MCM location ($\chi^2 = 124.75$, $p < 0.001$). Respondents who reported frequent outages were significantly more likely to prioritize low-income neighborhoods and residential areas, while placing less emphasis on critical infrastructure. This suggests that frequent disruptions heighten the perceived need for localized community-level energy resilience.

In contrast, the analysis of outage duration revealed an inverse pattern ($\chi^2 = 96.62$, $p < 0.001$). Households that experienced prolonged outages were more likely to prioritize public utility facilities and critical infrastructure, and less likely to choose low-income communities or industrial areas. These findings imply that different types of outage experiences shape distinct resilience priorities, with frequent disruptions motivating a focus on residential equity and prolonged outages prompting concern for systemic continuity and infrastructure reliability.

DISCUSSION

Results of this analysis echo previous studies that found perceived benefits and risks, and socioeconomic demographic factors affect an individual's attitude toward clean energy technologies. Consistent with existing literature, we find that perceived benefits, such as reliable power supply, outage reduction, disaster response, and energy cost savings, increase public support for MCMs. Conversely, belief in MCMs causing power supply disruptions lowers the acceptance intention. Beyond individual perceptions, social dynamics strongly influence how communities engage with MCM initiatives. Trust in implementing agencies serves as a key gatekeeper; when utilities and local governments are viewed as credible and transparent, residents are more likely to perceive MCMs as safe, fair, and worthwhile.[63] Social cohesion, including norms of mutual support and local networks, can facilitate participation in shared-energy programs and strengthen collective action.[64] Conversely, perceived inequities in siting, costs, or benefits may trigger opposition or disengagement, particularly in communities with a history of unequal energy access.[42] Ensuring meaningful involvement in decision-making and equitable distribution of resilience benefits can therefore enhance social legitimacy and sustain long-term support.[40,65] Additionally, this study finds that political ideology and outage experience are significant predictors of MCM acceptance, reinforcing existing evidence. However, our data indicate that male respondents report a higher likelihood of accepting MCMs than female respondents, which contrasts with prior research suggesting that women, particularly well-educated women, often play a leading role in clean energy.[66,67]

Geographic location priority and financial incentives

Translating public support into equitable deployment of MCMs requires that siting decisions reflect both power outage risk and community preferences. Our findings indicate that critical infrastructure and low-income neighborhoods are most frequently identified as priority locations. However, preferences vary based on residents' outage experiences. Individuals who experience frequent outages tend to prioritize residential and low-income areas, while those who endure longer outages are more likely to prioritize public utilities and critical infrastructure. These patterns underscore the need for MCM deployment strategies that integrate both objective outage data and community input to ensure equitable and effective outcomes. To foster trust and local ownership, policymakers should adopt participatory planning processes that enable communities to influence site selection based on lived experiences. Deployment efforts should begin in high-risk zones identified by both outage frequency and community feedback, such as affordable housing complexes, mobile home parks, and healthcare facilities in densely populated districts. Municipalities should also establish local

Table 2. Results of hierarchical linear regression models on MCM acceptance

Variables	Model 1	Model 2	Model 3	Model 4	Model 5
Step 1 Socio-demographics					
Income	0.016	0.015	0.015	0.008	0.013
Political views	−0.840***	−0.085***	−0.043**	−0.028*	−0.024
Gender (male = 1)	0.136***	0.141***	0.119***	0.124***	0.126***
Race (White = 1)	0.008	−0.001	0.048	0.036	0.049
Step 2 Power resilience					
Power outage frequency	–	0.036**	0.020*	0.024*	0.014
Power outage duration	–	0.017	0.010	0.009	0.016
Step 3 Power-related MCM					
Reliable power supply	–	–	0.505***	0.394***	0.349***
Power outage reduction	–	–	0.146***	0.082***	0.054*
Power supply disruption	–	–	−0.057**	−0.036	−0.056**
Step 4 Safety-related MCM perceptions					
Rapid responses to disaster	–	–	–	0.231***	0.220***
Battery disposal problem	–	–	–	−0.023	−0.007
Step 5 Finance-related MCM perceptions					
Lower energy cost	–	–	–	–	0.124***
Increase financial burden	–	–	–	–	0.004
R^2	0.023	0.031	0.391	0.424	0.438
ΔR^2	0.023	0.008	0.360	0.033	0.014
ΔF	9.550	6.523	319.050	46.441	20.138
–	***	**	***	***	***

*$p < 0.050$. **$p < 0.01$ ***$p < 0.001$. Political views (1 = liberal, 5 = conservative), Gender (male = 1), and white vs. non-white (white = 1) were coded as dummy variables. Education category "High school or lower" includes respondents with or without a high school diploma, which could introduce additional variance in perceptions.

energy advisory boards with resident representation to co-design deployment priorities and decision-making frameworks.

In community-led or utility-managed microgrid initiatives, financial incentives must be carefully aligned with governance and financing structures.[68] Rather than relying on direct household contributions, MCMs can be financed through public-sector instruments that equitably distribute costs. Resilience bonds, which monetize the value of avoided disaster losses, offer a promising model for front-loading investment in resilient energy infrastructure.[69,70] Additionally, municipal utility rebate programs and federal sources such as FEMA's Building Resilient Infrastructure and Communities (BRIC) initiative can also provide crucial support without imposing upfront costs on individual households.[71] BRIC has already funded numerous emergency and backup generators for disaster mitigation and could similarly support MCM projects. In 2024, FEMA further announced that three grant programs, BRIC, the Public Assistance program, and the Hazard Mitigation Grant Program, may be used to fund net-zero energy projects to rebuild critical infrastructure after disasters. Where microgrids are eligible, state and local governments and tribal nations can leverage these resources to help finance MCM deployment. Integrating MCMs into disaster response and recovery programs would align with FEMA's resilience goals by promoting cleaner and more flexible backup solutions. This alignment could also facilitate collaboration between utilities, local governments, and federal agencies to support scalable deployment. MCM projects can additionally draw on community solar funding models, using public-private partnerships with participant subscriptions. In such models, municipal utilities finance upfront costs using federal and state support, while subscribers pay a monthly fee for their share of resilient capacity. These mechanisms not only enhance the feasibility of MCMs but also advance energy equity by expanding access to backup power across income groups. To strengthen transparency and community engagement, all financing approaches should be paired with clear, accessible communication about long-term economic and resilience benefits for both residents and local governments.

Local government regulations and utility rules should be updated to explicitly accommodate microgrid deployment. Some cities have rules that limit the installation of renewable energy, while most zoning codes don't have renewable infrastructure included. Cities can add clear definitions for microgrids and Battery Energy Storage Systems (BESS), permit them as accessory uses across most districts, authorize cross-parcel feeders via easements or right-of-way permits, and create resilience overlay zones with expedited, ministerial permitting. Regulators should update the technical rules (e.g., how things connect to the grid) and the economic rules (e.g., how customers are charged and rewarded), so that it's easier and faster to connect distributed resources like rooftop solar, batteries, microgrid, etc., and so that the grid properly values and supports resilience and

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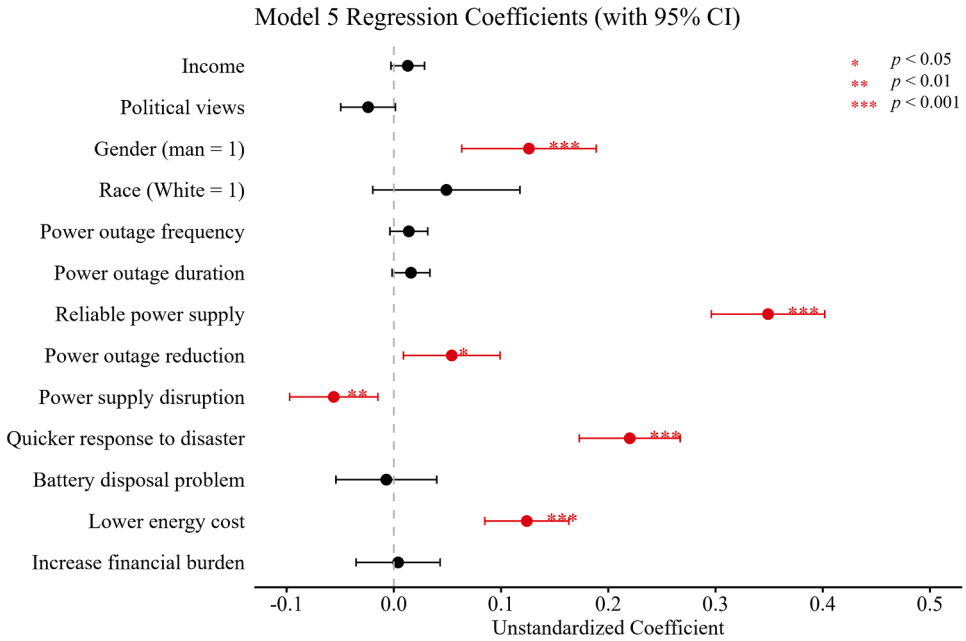

Figure 5. Results of model 5 (final model): estimated coefficients and confidence intervals for MCM acceptance
Significant predictors are highlighted in red, while nonsignificant predictors appear in black.

community-serving energy systems. On funding, MCMs can adopt a community-subscription model: the municipal or cooperative utility finances the project (braiding federal/state support and direct-pay credits), while subscribers pay a monthly fee for a share of resilient capacity, with discounted tiers for renters and income-qualified households. Pairing on-bill financing or inclusive utility investment with community benefit agreements ensures participation from renters and low-income customers and builds durable local benefits.

Perceived benefits and risks

Among the perceived benefits of MCM, expectations of lower energy costs significantly increased support for acceptance, suggesting that financial savings are a meaningful motivator. Additionally, reliable power supply, rapid disaster responses, and reduced outage frequency remained strong positive predictors, reinforcing the importance of resilience-related benefits. In contrast, perceived risks, including concerns about increased financial burdens and battery disposal issues, did not significantly deter acceptance. Interestingly, the perception of increased power disruptions, previously non-significant, regained statistical importance when financial perceptions were introduced, suggesting a more nuanced risk-benefit calculation among respondents. In contrast, battery fire safety is not a significant concern that affects support for microgrids. Although other technical risks were not tested in this study, such as grid interconnection, maintenance, and scalability, they are important practical challenges in microgrid deployment. Additionally, several technical and operational factors may constrain the large-scale deployment of MCMs. Integrating MCMs with existing distribution networks requires compatible interconnection standards, synchronization control, and safety protection schemes, which remain under development in many regions. Effective MCMs should also address operational issues to ensure voltage and frequency stability, power quality, and cybersecurity.[72] Regular maintenance and life cycle management of modular components, such as storage systems (battery degradation), power converters, panel cleaning, and mobile transport units, pose additional financial and logistical challenges, particularly for resource-limited municipalities.[73] Scalability also presents an obstacle: a design optimized for a dense urban block may not perform efficiently in dispersed rural settings due to variations in load profiles, communication infrastructure, and local regulations. Addressing these challenges through standardized technical guidelines, workforce training, and pilot-scale testing will be essential to ensure reliable, safe, and equitable MCM deployment.

MCMs directly contribute to ongoing U.S. energy-transition and climate-resilience goals by integrating cleaner distributed resources, providing reliable power during outages, and supporting decarbonization targets for 2035 and 2050.[74] Their mobility enables rapid deployment to critical facilities and neighborhoods after extreme weather events, aligning with national preparedness and hazard-mitigation frameworks that emphasize maintaining lifeline services and accelerating community recovery. When paired with equitable financing strategies, MCMs can extend resilience benefits to underserved households and community facilities, ensuring that clean-energy investments support both decarbonization and social equity objectives.

Resilience experiences, backup systems, and microgrid deployment

Our findings highlight the central role of resilience concerns in shaping public support for MCMs and preferences for their

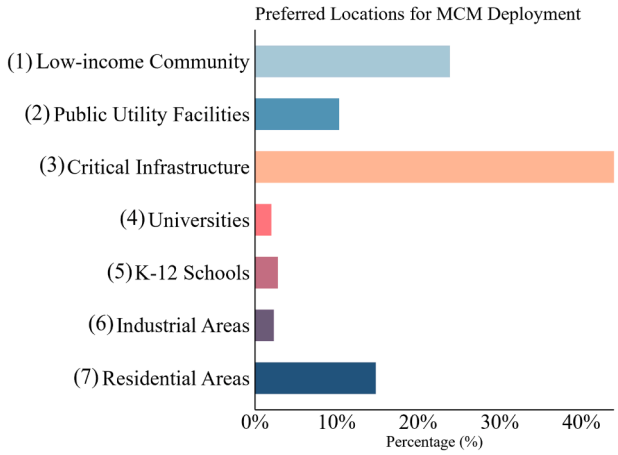

Figure 6. Distributions of preferred location for MCM

Bar plot for distributions of preferred location for MCM. Critical infrastructure (44.0%) received the greatest support, followed by low-income communities (23.8%), residential areas (14.8%), public-utility facilities (10.3%), K-12 schools (2.8%), industrial areas (2.3%), and universities (2.0%). Percentages sum to 100% after rounding.

deployment. The strongest predictors of MCM acceptance are perceptions of improved power reliability and disaster response, followed by expectations of lower energy costs. This pattern highlights the role of perceived benefits in driving community support for energy innovations.[75] Therefore, communication strategies should focus on enhancing public understanding of how MCMs improve power reliability during disruptions and contribute to disaster preparedness. Public education campaigns should explicitly link MCMs to practical benefits, such as continuity of critical services during storms or blackouts. Additionally, financial incentives, such as rebates or subsidies, should be coupled with clear cost-benefit narratives that emphasize long-term savings.

While the frequency of power outages significantly predicts openness to MCMs, the duration of the longest outage does not. This result suggests that frequent, even brief, disruptions may be more effective in raising public awareness and motivating interest in resilience solutions. Utilities should therefore target outreach in areas with high outage frequency, where residents may be particularly receptive to MCM implementation. Notably, higher-income respondents in our sample reported longer outage duration, likely reflecting suburban locations with less resilient infrastructure. The result points to the importance of addressing geographic vulnerabilities in MCM policy and planning. Integrating outage frequency and restoration time metrics into utility vulnerability assessments can better inform equitable site selection for MCM pilots. Suburban and rural areas in storm-prone or wildfire-risk regions may warrant special consideration due to their extended restoration timelines and heightened vulnerability.

Although ownership of backup generators did not significantly influence MCM acceptance, it was more common among high-income and politically conservative individuals. Household backup generator ownership patterns likely reflect geographic clustering. Rural and some suburban households, where restoration time is often longer, and grid redundancy is lower, show higher uptake of home backup generators.[76] This spatial variation can partially confound the apparent income effect, indicating that future analyses should model geographic (rural/urban/suburban) and socioeconomic factors jointly. This result indicates that home backup systems may not compete with community-scale microgrids but instead represent parallel approaches to energy resilience. Policymakers should frame MCMs as complementary to household energy systems and promote hybrid models where both coexist. Incentive structures, such as municipal resilience credits, state-level energy rebates, or tax benefits, could be designed to encourage coordination between MCMs and home systems. Outreach efforts should emphasize that MCMs enhance, rather than replace, individual resilience investments. In suburban areas with high generator ownership (e.g., gated communities or newer developments), integrated models could position MCMs as shared resource hubs that improve fuel efficiency and extend operation during long outages.

Social-demographic drivers of microgrid acceptance

Our results indicate that gender and political orientation significantly influence MCM acceptance intention, while traditional demographics such as income and race do not. While lower-income and minority households do face disproportionate energy burdens,[77,78] recent studies highlight the importance of gender and political ideology in shaping energy attitudes and behaviors.[79,80] The perception that women might be less interested in accepting microgrids or other renewable technologies often reflects systemic barriers. Prior work shows that gender norms, unequal access to information and finance, and institutional arrangements shape women's engagement with energy-related technologies, with men frequently retaining primary decision-making authority over household appliances and energy technologies.[81,82] In the renewable energy sector more broadly, gender bias, unequal access to Science, Technology, Engineering, and Mathematics (STEM) education and technical training, and weak organizational support continue to restrict women's participation, especially in technical and leadership roles.[83,84] Additionally, women also face greater financial constraints and heightened perceptions of financial risk in capital-intensive "citizen energy" investments, contributing to their underrepresentation in community renewable projects.[84,85]

Our finding suggests liberals generally have higher or more acceptance of MCMs than conservatives in our sample. Political ideology influences MCM adoption because it shapes how people understand climate risks, disasters, how much they trust government and utilities, and whether they prefer individually owned, market-based solutions or collective, publicly coordinated resilience infrastructure. For example, liberals are more supportive of collective, publicly funded, or regulated interventions (public investments, standards, resilience programs).[86,87] Accordingly, policy initiatives promoting MCM deployment should adopt intersectional frameworks that explicitly account for gender differences and ideological orientations rather than relying solely on economic or racial indicators. Targeted outreach can improve effectiveness: for example, campaigns aimed at women-headed households, particularly in urban

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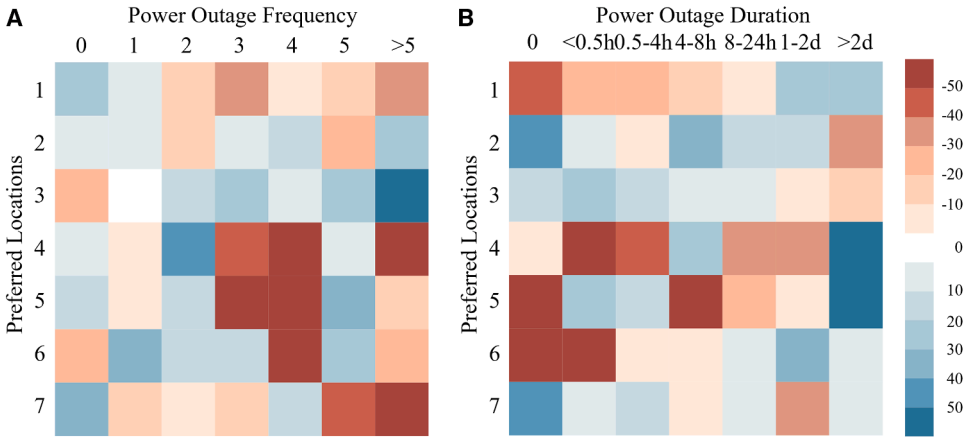

Figure 7. Differences in percentages of selecting a preferred location by power outage frequency and duration

Stacked bar plot for income groups with varying power outage duration. (A and B) represent the cross table of the differences from the average distribution for power outage frequency, duration, and preferred location. Each cell represents the percentage difference between the selection of a preferred location under the given column's power outage frequency/duration and the overall percentage of selecting that location in a. White indicates no difference, while darker blue represents a higher percentage than the total percentage, and darker green represents a lower percentage.

areas, should emphasize family safety, continuity of healthcare, and fair cost sharing; outreach in conservative-leaning suburban communities may resonate more with frames of energy independence and local control. Although direct out-of-pocket costs were not major deterrents, anticipated economic benefits significantly motivated acceptance. Therefore, educational materials should foreground cost savings and resilience benefits, with loss-avoidance scenarios framing where appropriate.[88,89] Moreover, messaging and incentives should differentiate between homeowners, who tend to value long-horizon resilience investments, and renters, who often perceive fewer direct benefits and may require landlord-mediated programs or portability features to meaningfully participate.[90–92]

Equitable access and affordability are critical to ensure MCM initiatives do not inadvertently reinforce existing energy inequities.[47] Renters often have limited control over household energy decisions and are frequently excluded from programs that are designed around property ownership.[93,94] If this structural limitation is not addressed, benefits such as improved outage protection or lower electricity costs may concentrate among homeowners or higher income neighborhoods.[47,95] Marginalized communities may also face barriers, including limited technical information, fewer opportunities for participation, and reduced influence in local decision-making. These barriers underscore the need for inclusive planning that actively incorporates equity considerations and addresses existing injustice.[42] Effective strategies include providing targeted financial assistance, developing rent-inclusive pricing models, and partnerships with trusted local community organizations to broaden participation.[65] Transparent governance sustained community engagement, and communication about benefits, costs, technological information and responsibilities on a regular basis can prevent new forms of inequality and help build public trust in microgrid implementation programs.[47]

The weakening effect of political ideology in later models suggests that support for MCMs may depend less on partisan identity and more on trust in institutions responsible for energy and infrastructure management.[47,63] In polarized contexts, ideological cues can initially shape perceptions of new technologies, but practical concerns such as reliability, cost, and local governance tend to override partisan differences.[96,97] Trust provides the mental foundation for generating team spirit and collaborative efforts that help overcome the political and cultural differences among community members.[98] Trust in infrastructure is important to the adoption of new complex technical systems, and lack of trust always impedes project deployment.[99] Building institutional transparency and emphasizing equitable community benefits can therefore reduce ideological resistance and strengthen public trust in MCM deployment.[42,50]

Social acceptance and community support for mobile community microgrids

Social acceptance of MCMs extends beyond general approval of new energy technologies to encompass community trust, perceived fairness, and flexibility in deployment and operation.[42,63] Because MCMs are designed to be mobile and temporary, acceptance depends not only on their technical reliability but also on how their movement and duration of deployment are managed at the community level.[100] Residents are more likely to support MCM projects when deployment timelines are transparent, decision processes are participatory, and power allocation is perceived as equitable.[100,101] Short- to medium-term deployments (e.g., several weeks to a few months following major outages) may be suitable for post-disaster recovery, whereas longer placements in vulnerable communities could serve as transitional resilience measures. Together, these social and temporal dynamics clarify what community support and social acceptance look like for MCMs in practice.

Limitations of the study

This study investigates key social-psychological factors (e.g., perceived risks and benefits), demographic characteristics,

Table 3. Summary of descriptive statistics of the sample

Variable	Observation	Mean	Std. dev.	Min	Max
Age	1,996	44.8	17.3292	18	89
Female	1,972	0.46	0.4982	0	1
Income ($k)	1,756	65.32	45.9126	12.5	15
Ideology	1,737	3.0	1.1991	1	5
Race/ethnicity	White: 63.2% Black: 16.2% Hispanic: 19.3% Asian: 5.8% Native: 2.7%		**Income ($k)**	Low-income: 46.7% Medium-income: 36.9% High-income: 16.4%	
Monthly Electricity Bill	<$50: 7.3% $50-$99: 26.6% $100-$149: 33.6% $150-$200: 20.0% >$200: 12.6%		**Education** High school or lower: 36.3% Associate's or some college: 27.3% Bachelor's degree: 23.1% Graduate degree: 13.3%		

Income is a categorical variable with 15 categories; Race/ethnicity allows multiple choices.
Ideology was coded by the 5-point scale; 1 = conservative, 5 = liberal. Gender was coded as a dummy variable.

and resilience-related variables associated with public support for MCMs in the U.S. While the findings offer valuable insights, several limitations warrant consideration. First, the survey did not collect ZIP code data, limiting our ability to examine spatial variation in MCM preferences or link public perceptions with local grid infrastructure, outage durations, or exposure to climate-related hazards. This absence constrains efforts to refine place-based strategies for microgrid deployment. Future research should incorporate geographic identifiers (e.g., ZIP code) to support spatial regression analyses and geographically weighted modeling. Because outage experiences were self-reported, our data may be affected by recall bias and subjective interpretation. Respondents who are more sensitive to service interruptions may overestimate outage duration or frequency, whereas others may underreport brief disruptions. Nonetheless, outage perceptions capture important aspects of household lived experience (e.g., fear of food spoilage, missed work, concerns about health-related devices), even when objective metrics are unavailable. In practice, researchers generally cannot directly observe each household's outage history, and access to granular utility outage records is often constrained by privacy, legal, or proprietary restrictions that vary across utilities. Future work should triangulate subjective reports with objective indicators where feasible, such as utility reliability metrics, neighborhood-level restoration logs, or aggregated smart-meter data.

Second, the survey did not measure housing tenure, whether respondents owned or rented their homes. Housing tenure likely influences perceived benefits of resilience investments and the decision-making autonomy relevant to adopting community-scale infrastructure. Future studies should explore how tenure status shapes support for MCMs and whether outreach or incentive programs should be tailored accordingly.

Third, another limitation is that this study examines behavioral intention and acceptance rather than observed adoption. Although intention is a strong predictor of future action, it may not fully capture the gap between stated support and actual participation. Future research should incorporate behavioral measures or follow-up data to assess how intentions translate into realized adoption outcomes.

Fourth, we acknowledge that our cross-sectional design limits causal inference and does not capture attitudinal change over time. Longitudinal approaches, particularly pre-post designs around outage events or MCM deployment, would provide clearer evidence on how perceptions evolve. In addition, experimental vignette (conjoint) designs that manipulate salient MCM attributes (e.g., reliability gains, cost, safety features) can directly test hypotheses and strengthen causal claims. Future research should incorporate these approaches to more rigorously evaluate the determinants of public support for MCMs.

Fifth, contrary to conventional assumptions, our findings indicate that higher-income households can experience frequent or prolonged power outages due to intersecting geographic, infrastructural, and behavioral factors. Higher-income residents tend to have greater expectations for uninterrupted power and are more likely to perceive outages as severe disruptions.[102] These behavioral dynamics can influence utility data, shape policy discourse, and amplify narratives about reliability challenges. The relationship between income and outage duration, therefore, requires further validation, particularly in the context of increasing extreme weather impacts. To better understand these dynamics, future research should use mixed methods and integrate real-time outage data to examine how income inequality shapes power disruption experiences. Overall, infrastructure vulnerability is shaped not only by physical exposure but also by systemic and perceptual factors, underscoring the need for more nuanced resilience strategies.

Finally, while this study offers a conceptual framework linking MCM deployment strategies to public risk and benefit perceptions and social-demographics factors, future research should expand the scope to include a broader range of social-psychological variables. Moreover, it is critical to acknowledge that key decisions regarding MCM implementation are made not by individual residents, but by local governments and utility providers. As such, future studies should integrate the perspectives,

priorities, and operational constraints of these institutional actors. A deeper examination of regulatory frameworks, utility governance structures, and equity considerations will be essential to inform practical, scalable, and socially just strategies for community microgrid deployment.

Conclusions

This study provides foundational evidence for researchers, policymakers, and industry stakeholders to better understand public attitudes toward MCMs. By identifying key resilience-related, socio-demographics, and perceptual drivers that shape acceptance, it offers empirically grounded guidance for deployment and policy design, while underscoring the need to validate these patterns across diverse communities and geographies. A persistent challenge is the disconnect between public preferences and institutional decision-making, since MCM siting and implementation are led primarily by local governments and utility providers rather than individual residents. Open, participatory governance structures that encourage civic participation, including regular public hearings and town hall meetings, structured community-engagement programs, and public-private partnerships, provide established channels to incorporate residents' opinions into routine institutional decision-making. Community co-production approaches can further align designs with local needs, strengthen legitimacy, and improve uptake. Advancing equitable and scalable MCM deployment will therefore require future research that directly engages utilities, planners, and regulators, and critically examines the regulatory, governance, and equity architectures shaping infrastructure choices. This work serves as a starting point, laying the groundwork for inclusive, and context-sensitive strategies for resilient energy systems.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Chien-fei Chen (chienfc@clemson.edu).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- The data contain human participant information and are therefore not publicly available to protect participant privacy and comply with human-subjects protocols.
- No custom code was generated for this study.
- The survey sample included 1,996 respondents, recruited via the Qualtrics online panel using quotas to approximate national representation by race/ethnicity, gender, and income.
- There are no clinical trials.
- Any additional information required to reanalyze the data reported in this article is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, J.X. and C-F C.; methodology, J.X. and C.-F. C.; investigation, C.-F. C. and Y.W.; data curation, Y.W.; formal analysis, J.X. and C.-F C.; writing – original, J.X. and C-F. C.; writing – review and editing, J.X., C.-F. C., and Y.W.; funding acquisition, C.-F. C., and Y.W.; supervision, C.-F. C.

DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work the authors used Grammarly to correct grammatical errors. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- METHOD DETAILS
  - Study participants details
  - Data analysis strategy
  - Home backup power system analysis
  - MCM location preference analysis
- QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Mobile Community Microgrid Acceptance	This study	Microgrid acceptable was based on the survey data and calculated by this study
Social demographic and social-psychological factors	This study	Demographics and social-psychological factors were based on the survey data and calculated by this study
Software and algorithms		
SPSS Statistics version 29.0	IBM	https://www.ibm.com/products/spss-statistics
Survey Design Platform	Qualtrics	https://www.qualtrics.com/en-gb/
Participant Recruitment	Dynata Online panel	https://www.dynata.com

METHOD DETAILS

Study participants details

This study used an online survey targeting adults who were fully or jointly responsible for paying their household energy bills. Respondents were first introduced to basic information about MCMs via a brief textual description, supported by real-world examples and a simple diagram. After viewing this material for at least 30 s, participants answered questions on their intention to adopt MCMs (our primary measure of MCM acceptance), followed by items capturing social-psychological factors, including attitudes toward microgrids and perceived risks and benefits of MCMs. Finally, participants reported their power outage experiences, preferred MCM locations, and demographic characteristics.

A nationally representative sample was recruited through Dynata's online panel, with quotas for age, gender, race, and income to ensure demographic balance. The final analytic sample comprised 1,996 valid responses. Respondents had an average age of 45 years; 46.0% identified as female, 63.2% as White, 16.2% as Black, and 19.3% as Hispanic. Income distribution in the sample included 46.7% low-income households (LIHs; income < $50k), 36.9% middle-income households (MIHs; $50k–$125k), and 16.4% high-income households (HIHs; > $125k). Reported monthly electricity bills ranged from under $50 (7.3%) to more than $200 (12.6%): 26.6% reported bills of $50–$99, 33.6% $100–$149, and 20.0% $150–$200. Descriptive statistics for the sample are summarized in Table 3. Political ideology was measured on a five-point Likert scale ranging from 1 (mostly liberal) to 5 (mostly conservative) and was treated as an interval variable in the regression analyses, following standard practice in social and energy behavior research.

Data analysis strategy

We followed a structured, multi-stage OLS regression analysis strategy. First, we computed descriptive statistics to summarize sample characteristics and key variable distributions. Next, we estimated a five-block hierarchical multiple regression to test hypotheses and determine whether socio-demographics, power-outage experiences, and social-psychological factors of MCMs predicted acceptance. The hierarchical approach enabled us to add conceptually related predictors in sequence and to examine both direct effects and the incremental variance explained by each block (ΔR^2). Additionally, we examined the influence of monthly electricity bill and found it was not a significant predictor of MCM acceptance (see Table S4 in Supplemental Material). Independent variables were entered in a theoretically grounded order reflecting expected causal pathways. In Step 1, socio-demographic variables served as baseline controls. Step 2 added outage-experience variables, consistent with evidence that more frequent or prolonged outages increase interest in resilience technologies. Steps 3–5 incorporated perception-based (i.e., social-psychological) variables related to MCMs. To determine the optimal entry order for these perception domains (power, safety, and finance), we first estimated separate models for each domain to gauge its standalone explanatory power. We then sequentially added each domain to the Step-2 baseline and compared changes in model fit and selected the sequence that maximized explanatory power and statistical significance.

As shown in Table S2, perceptions related to power reliability and supply disruption emerged as the strongest individual variables and were therefore entered first in Step 3. This aligns with prior research emphasizing the central role of perceived power reliability in energy-related decision-making.[103] In Step 4, safety-related perceptions, including views on rapid disaster response and risks associated with MCM technology, further improved model performance. Finally, financial perceptions were added in Step 5, contributing a modest yet meaningful explanatory power.

Home backup power system analysis

To examine the relationship between home backup generator ownership and MCM acceptance intentions, we first estimated a linear regression model including only generator ownership and MCM acceptance. We then estimated a second model that additionally controlled for power outage experiences and socio-demographic characteristics. This model included only socio-demographic and outage-experience variables and excluded MCM social-psychological (perception) variables, as those were examined in separate models.

MCM location preference analysis

To assess residents preferred the primary location of MCM deployment, the survey included a single-choice question: "*Suppose your city or town is considering installing a microgrid to reduce the risks of power outages. In your opinion, what would be the best primary location for the microgrid to benefit your community the most*?" Response options included: low-income communities, public utility facilities, critical infrastructure, universities, K–12 schools, industrial zones, residential areas, other locations, and not sure. Responses marked as "other locations" were reviewed and, when appropriate, recoded to one of the predefined categories. Broad or non-specific answers were considered invalid and coded as missing, resulting in the exclusion of 159 responses.

To examine how power outage experiences shape MCM location preferences, we conducted Pearson Chi-square tests and ANOVA to assess the relationships between outage frequency, outage duration, and the preferred MCM location. Cross-tabulations were created to compute Chi-square statistics, and column percentages were analyzed to identify shifts in preferred location across different outage experience groups.

$$Change(i,j) = P_{col}(i,j) - P_{total}(i)/P_{total}(i) \quad \text{(Equation 1)}$$

Where $P_{col}(i,j)$ is the proportion of location i within outage frequency j, $P_{total}(i)$ is the overall proportion of location i across all outages. The result represents the percentage increase or decreases in preference for that location at a given outage level.

QUANTIFICATION AND STATISTICAL ANALYSIS

All statistical analyses were conducted using SPSS version 29.0. Multicollinearity in the regression models was evaluated using Variance Inflation Factors (VIF). All statistical tests were two-tailed. Relationships between continuous variables were analyzed using OLS multiple linear regression. To examine the predictors of MCM acceptance based on our hypotheses, we estimated a five-block hierarchical multiple regression analysis to assess the incremental variance explained by socio-demographics, power outage experiences, and social-psychological factors (including perceived risks and benefits, attitudes etc.). For comparisons of outage duration across income groups, we used one-way Analysis of Variance (ANOVA), followed by Tukey's post-hoc tests. Associations between categorical variables were assessed using Pearson Chi-square tests. Descriptive statistics were reported as mean ± standard deviation (*SD*) and regression results were presented as standardized regression coefficients (β) with 95% confidence intervals (*CI*). A p-value < 0.05 was considered statistically significant for all analyses.